



Cassidy MacPherson

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EDUCATION

UBC Engineering Physics - Embedded Systems Specialization

Expected Graduation – April, 2027

Engineering Physics is the most competitive and academically challenging engineering program at UBC. It is a 5 year program that combines mechatronics engineering with honors-level applied math and physics.

SKILLS

SW Languages	C, C++, Python, MATLAB, Java, Javascript, Assembly (x86, ARM).
SW Tools	ROS, OpenCV, Tensorflow, Scikit, NumPy, Pandas, FreeRTOS, Ubuntu, Gazebo, Git.
Electrical	Simulink/Simscape, LTSpice, Altium, PCB Assembly, Verilog, VHDL, Quartus, Modelsim.
Mechanical	Solidworks, Fusion360, 3D Printing, Laser Cutting, Metalworking, Woodworking.

DESIGN & WORK EXPERIENCE

Electrical Engineering Intern - Gentiam Electronics

May, 2025 - Present

- Contributed to the design of an electric powertrain model in Simulink/Simscape.
- Developed MATLAB scripts to automate testing of control algorithms and powertrain systems.
- Developed software for real-time sensor data acquisition and serial communication (CAN, I2C, UART) within the powertrain system.

Self-Driving Detective Robot

- Programmed an autonomous robot to obey traffic laws and avoid obstacles in a Gazebo environment.
- Designed and trained a CNN-based clue reader to read sign information from real-time camera feeds.
- Developed ROS nodes to process camera input, publish velocity commands, submit clueboard data, and respawn the robot through service calls.

RL Underactuated Watercraft

- Designed five solar-powered RC boats to autonomously drive between docking stations in a large tub.
- Implemented reinforcement learning algorithms to train boats to avoid collisions during navigation and maintain a stable stationary position when docked, even in windy conditions.
- Developed ROS nodes to manage communication, control, and sensor integration for the fleet.

Autonomous Cooking Robots

- Designed two autonomous robots for a real-life recreation of the video game Overcooked.
- Developed closed-loop control algorithms on an ESP32 microcontroller to interpret IMU data and execute precise motion control for both the chassis and arm.
- Implemented FreeRTOS to manage concurrent tasks and handle sensor-driven interrupts.
- Established a Wi-Fi-based communication between two ESP32 units to synchronize robot behaviors.